

Trapping Lemma for the Lorenz System

Let

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(r - z) - y, \\ \dot{z} &= xy - bz.\end{aligned}$$

Lemma. *There exists a bounded ellipsoid E that is a trapping region for the Lorenz flow. Equivalently, the time-one map T of the Lorenz flow satisfies*

$$T(E) \subset \text{int}(E).$$

Sketch of the Proof

- ① Lyapunov function. Define

$$V(x, y, z) = rx^2 + \sigma y^2 + \sigma(z - 2r)^2.$$

Along solutions of the Lorenz system,

$$\dot{V} = -2\sigma(rx^2 + y^2 + bz^2 - 2brz).$$

- ② Negativity outside a large ellipsoid.

The quadratic form $Q(x, y, z) = rx^2 + y^2 + bz^2 - 2brz$ is positive definite away from a bounded region. Therefore

$$\dot{V} < 0 \quad \text{outside a sufficiently large ellipsoid.}$$

- ③ Construct the sets D and E .

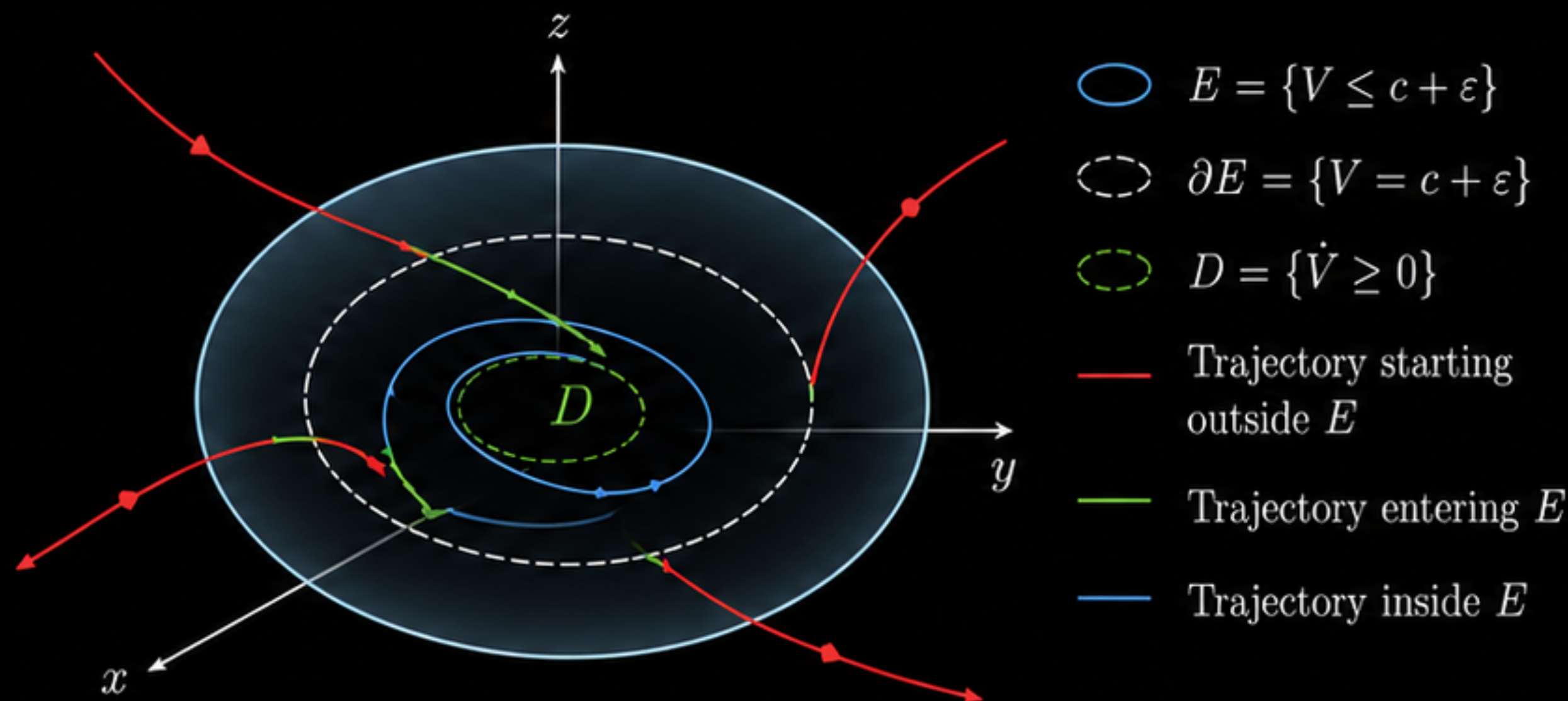
Let $D = \{\dot{V} \geq 0\}$. Then D is bounded (hence compact).

Let $c = \max_D V$ and $E = \{V \leq c + \varepsilon\}$

for some $\varepsilon > 0$. Then E is a bounded ellipsoid containing D .

- ④ Properties of E .

- **Outside E :** $\dot{V} < 0$. Hence $V(t)$ strictly decreases along any trajectory that starts outside E .
- Thus every trajectory enters E in finite time.
- On the boundary $\partial E = \{V = c + \varepsilon\}$ we have $\dot{V} < 0$, so the vector field points strictly inward. Therefore trajectories **cannot leave** E once inside.



$\Rightarrow$ Therefore E is a trapping region for the Lorenz flow. Equivalently, $T(E) \subset \text{int}(E)$.